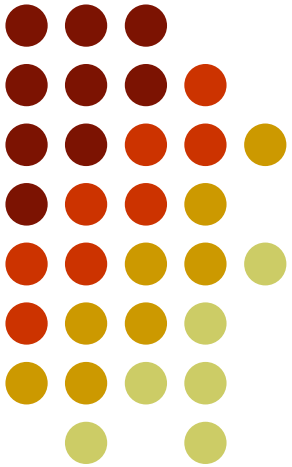
Properties of Stock Options

Chapter 10



Notation

- c : European call option price
- p : European put option price
- S_0 : Stock price today
- K : Strike price
- T : Life of option
- σ: Volatility of stock price
- C : American Call option price
- P : American Put option price
- S_T :Stock price at option maturity
- D : Present value of dividends during option's life
- r : Risk-free rate for maturity T with cont. comp.

Effect of Variables on Option Pricing (Table 10.1, page 233)



Variable	c	p	C	P
S_0	+	−	+	−
K	−	+	−	+
T	?	?	+	+
σ	+	+	+	+
r	+	−	+	−
D	−	+	−	+

Assumptions

- There are no transactions costs
- All trading profits are subject to the same tax rate
- Borrowing and lending are possible at the risk-free interest rate

American vs European Options



An American option is worth at least as much as the corresponding European option

$$C \geq c$$

$$P \geq p$$

Upper Bounds

- Call:

$$c \leq S_0,\ C \leq S_0$$

- Put:
  - American put: $P \leq K$
  - European put: $p \leq Ke^{-rT}$

Calls: An Arbitrage Opportunity?



- Suppose that

  $c = 3$ $\qquad$ $S_0 = 20$

  $T = 1$ $\qquad$ $r = 10\%$

  $K = 18$ $\qquad$ $D = 0$

- Is there an arbitrage opportunity?

Lower Bound for European Call Option Prices; No Dividends (Equation 10.4, page 238)

Consider two portfolios:

Portfolio A: one European call option plus a zero-coupon bond that provides a payoff of K at time T

Portfolio B: one share of the stock

$$c \geq \max(S_0 - Ke^{-rT}, 0)$$

Puts: An Arbitrage Opportunity?



- Suppose that

p = 1 $\quad S_0$ = 37

T = 0.5 $\quad r$ = 5%

K = 40 $\quad D$ = 0

- Is there an arbitrage opportunity?

Lower Bound for European Put Prices; No Dividends

(Equation 10.5, page 240)



Consider two portfolios:

Portfolio C: one European put option plus one share

Portfolio D: an amount of cash equal to Ke^{-rT} (or equivalently a zero-coupon bond paying off K at time T)

$$p \geq \max(Ke^{-rT} - S_0, 0)$$

Put-Call Parity; No Dividends



- Consider the following 2 portfolios considered before:
  - Portfolio A: European call on a stock + zero-coupon bond that pays K at time T
  - Portfolio C: European put on the stock + the stock

Values of Portfolios

		$S_T > K$	$S_T < K$
Portfolio A	Call option	$S_T - K$	0
	Zero-coupon bond	K	K
	Total	S_T	K
Portfolio C	Put Option	0	$K - S_T$
	Share	S_T	S_T
	Total	S_T	K

The Put-Call Parity Result (Equation 10.6, page 241)



- Both are worth max(S_T , K) at the maturity of the options
- They must therefore be worth the same today. This means that

$$c + Ke^{-rT} = p + S_0$$

Arbitrage Opportunities



- Suppose that

  c = 3 S_0 = 31

  T = 0.25 r = 10%

  K =30 D = 0

- What are the arbitrage possibilities when

  p = 2.25 ?

  p = 1 ?

Arbitrage Opportunities (con't)

Three-month put price = $2.25	***Three-month put price = $1***
Action now:	*Action now*:
Buy call for $3	Borrow $29 for 3 months
Short put to realize $2.25	Short call to realize $3
Short the stock to realize $31	Buy put for $1
Invest $30.25 for 3 months	Buy the stock for $31
Action in 3 months if $S_T > 30$:	*Action in 3 months if* $S_T > 30$:
Receive $31.02 from investment	Call exercised: sell stock for $30
Exercise call to buy stock for $30	Use $29.73 to repay loan
Net profit = $1.02	Net profit = $0.27
Action in 3 months if $S_T < 30$:	*Action in 3 months if* $S_T < 30$:
Receive $31.02 from investment	Exercise put to sell stock for $30
Put exercised: buy stock for $30	Use $29.73 to repay loan
Net profit = $1.02	Net profit = $0.27

American Options

- Put-call parity holds only for European options
- When there are no dividends, it can be proved that

$$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$$

Early Exercise



- Usually there is some chance that an American option will be exercised early
- An exception is an American call on a non-dividend paying stock, which should never be exercised early

An Extreme Situation

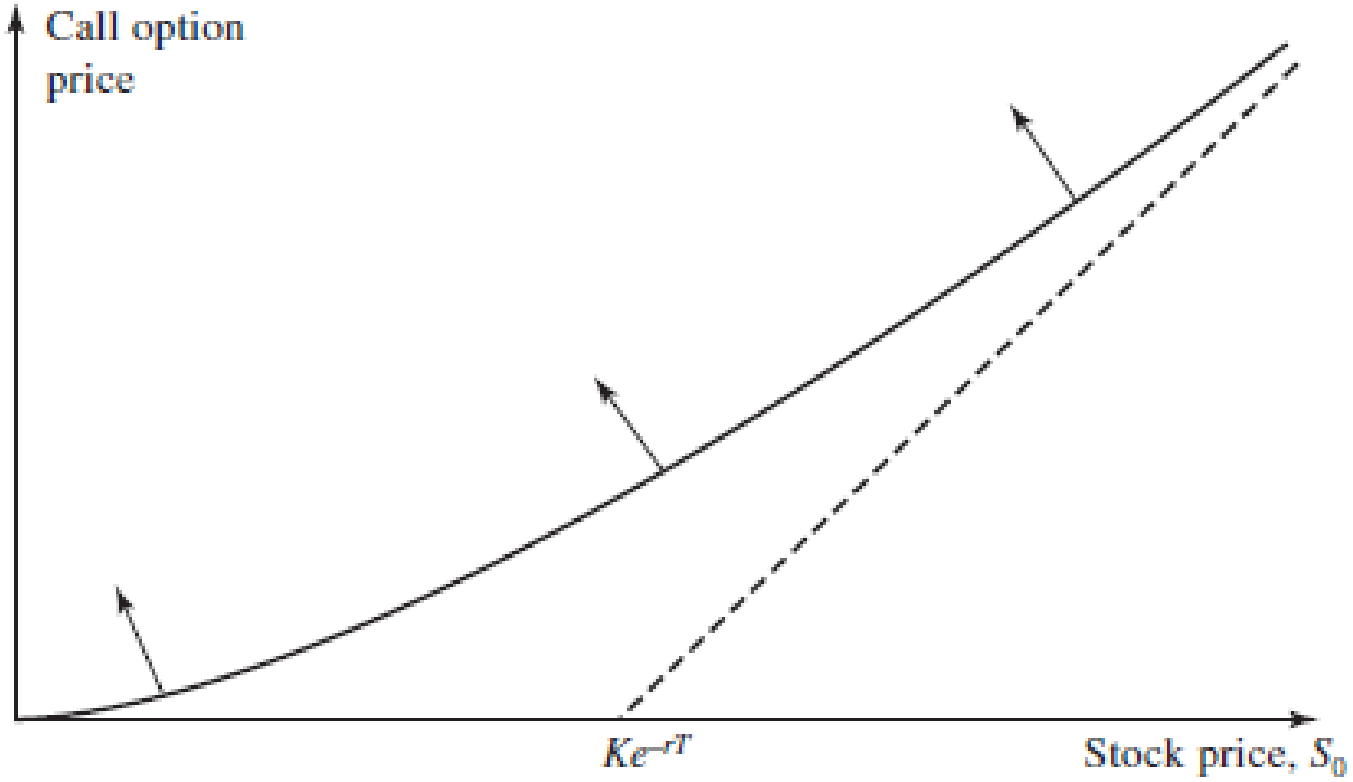


- For an American call option:

  $$S_0 = 100;\ T = 0.25;\ K = 60;\ D = 0$$

  Should you exercise immediately?
- What should you do if
  - You want to hold the stock for the next 3 months?
  - You do not feel that the stock is worth holding for the next 3 months?

Price Variation of a Call option on a Non-Dividend-Paying Stock

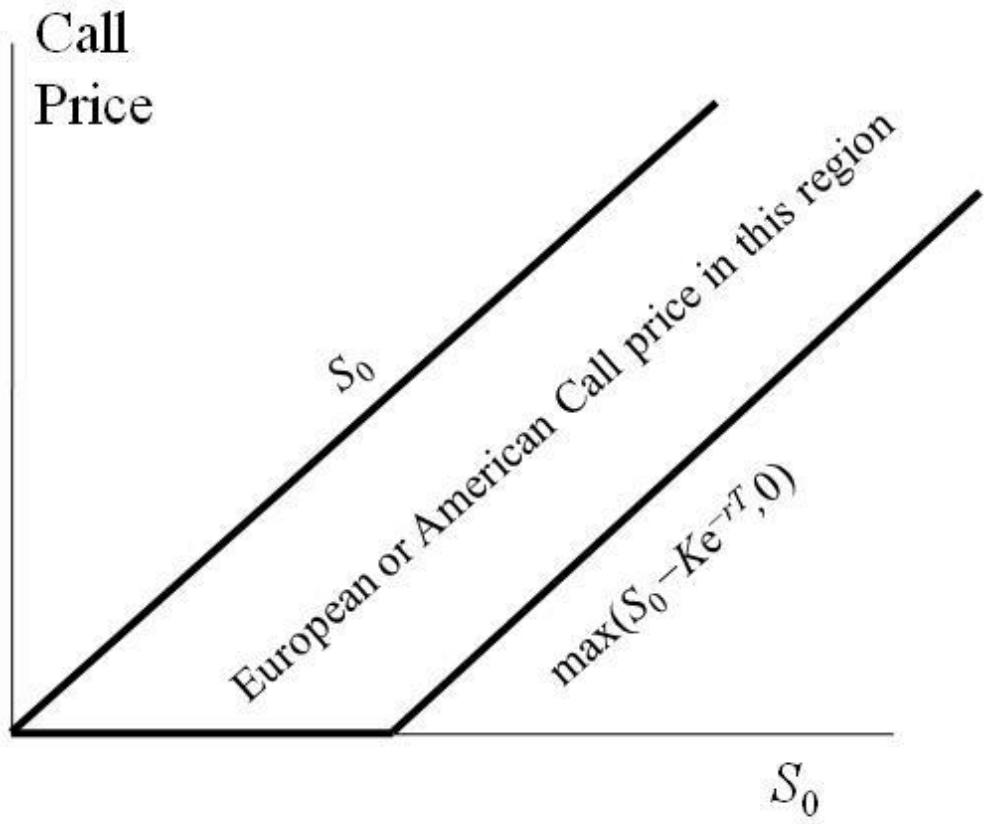


Reasons For Not Exercising a Call Early (No Dividends)



- No income is sacrificed
- You delay paying the strike price
- Holding the call provides insurance against stock price falling below strike price

Bounds for European or American Call Options (No Dividends)



Should Puts Be Exercised Early ?

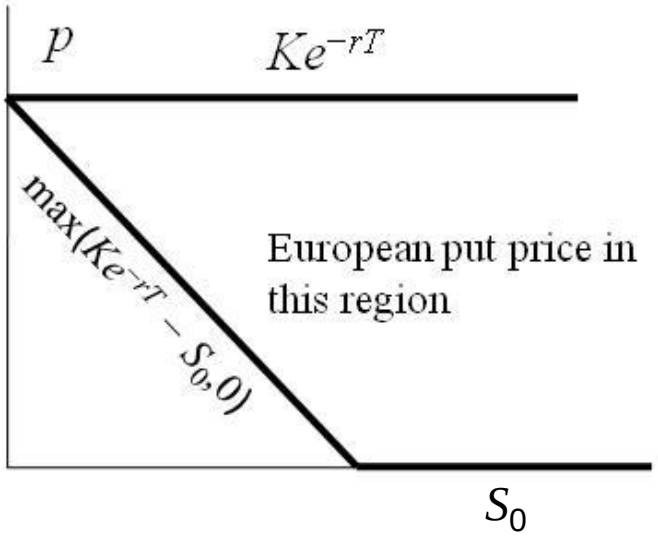


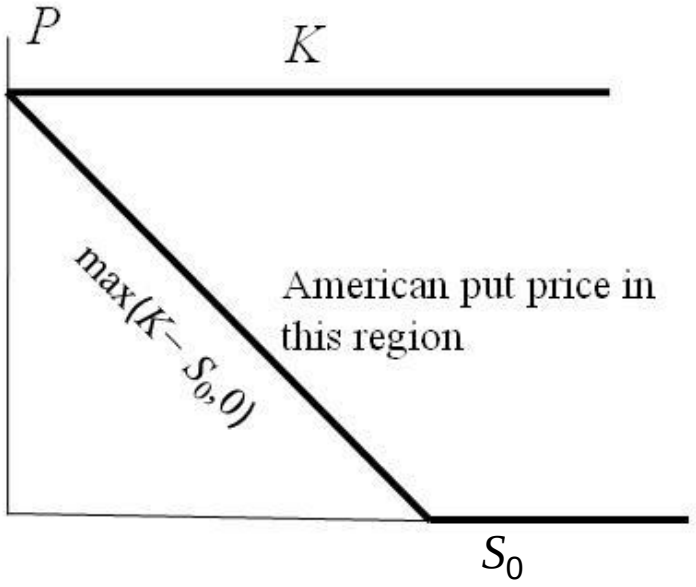
Are there any advantages to exercising an American put when

S_0 = 60; T = 0.25; r=10%

K = 100; D = 0

Bounds for European and American Put Options (No Dividends)





Impact of Dividends on Lower Bounds of Call Option Prices

(Equations 10.8 page 248)

Redefine portfolios A and B:

Portfolio A: one European call option plus an amount of cash equal to $D + Ke^{-rT}$

Portfolio B: one share

$$c \geq \max(S_0 - D - Ke^{-rT}, 0)$$

Impact of Dividends on Lower Bounds of Put Option Prices

(Equations 10.9 page 249)



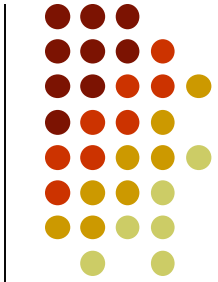
Redefine portfolios C and D:

Portfolio C: one European put option plus one share

Portfolio D: an amount of cash equal to $D + Ke^{-rT}$

$$p \geq \max(D + Ke^{-rT} - S_0, 0)$$

Extensions of Put-Call Parity



- American options; $D = 0$

  $S_0 - K < C - P < S_0 - Ke^{-rT}$

  Equation 10.7 p. 243

- European options; $D > 0$

  $c + D + Ke^{-rT} = p + S_0$

  Equation 10.10 p. 249

- American options; $D > 0$

  $S_0 - D - K < C - P < S_0 - Ke^{-rT}$

  Equation 10.11 p. 249

Summary

- Effect of variables on option pricing
  - Table on P3
- Properties of stock options
  - Three assumptions: no transaction costs, same tax rate, borrowing and lending at r_f
  - American vs European options
    - $C \geq c$, $P \geq p$

Summary

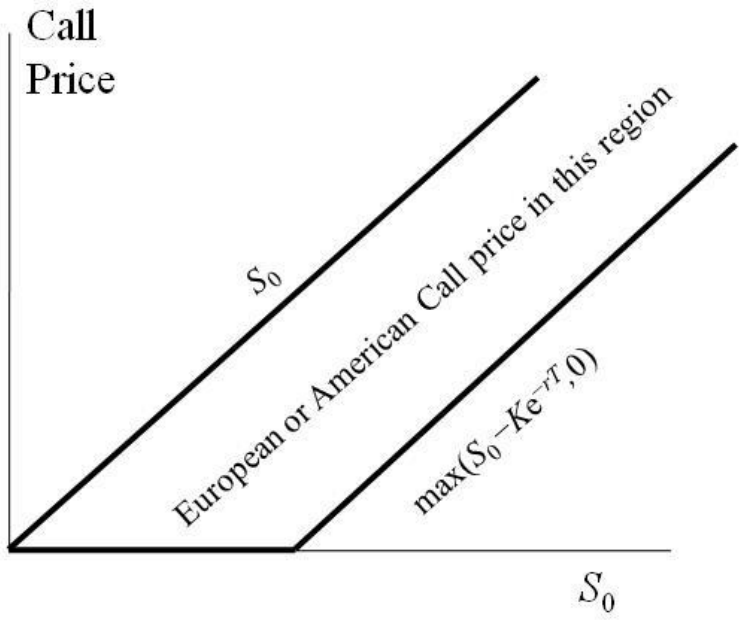
- Properties on stock options (cont'd)
  - Upper bonds
    - Call: $c \le S_0$, $C \le S_0$
    - Put: $P \le K$ (American), $p \le Ke^{-rT}$ (European)
  - Lower bond
    - European call, D = 0: $c \ge \max(S_0 - Ke^{-rT}, 0)$
    - European put, D = 0: $p \ge \max(Ke^{-rT} - S_0, 0)$
    - European call, D > 0: $c \ge \max(S_0 - D - Ke^{-rT}, 0)$
    - European put, D > 0: $p \ge \max(D + Ke^{-rT} - S_0, 0)$

Summary

- Put-call parity
  - European options, D = 0
    - $c + Ke^{-rT} = p + S_0$
  - European options, D > 0
    - $c + D + Ke^{-rT} = p + S_0$
  - American options, D = 0
    - $S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$
  - American options, D > 0
    - $S_0 - D - K < C - P < S_0 - Ke^{-rT}$

Summary

- Early exercise
  - American call on a non-dividend paying stock: should never be exercised early
  - Bounds for European or American call options (no dividends)



Summary

- Early exercise (cont'd)
  - American put: should always be exercised early if it is sufficiently deep in the money
  - Bounds for European and American put options (no dividends)

